



## *Java SE 8 Programming Language*

# **Training Assignments**

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**Contents**

|                                                                                             |   |
|---------------------------------------------------------------------------------------------|---|
| Long Assignment 1 – Option 1: Advanced OOP, Generic and Collection, Stream and Filter ..... | 4 |
| Objectives: .....                                                                           | 4 |
| Working Environments .....                                                                  | 4 |
| Assignment Descriptions .....                                                               | 4 |
| Validation Rules .....                                                                      | 5 |
| Functional Requirements .....                                                               | 5 |
| User Interface Requirements .....                                                           | 5 |



CODE: JPL.L.A101  
TYPE: LONG  
LOC:  
DURATION: 180 MINUTES

## Long Assignment 1 – Option 1: Advanced OOP, Generic and Collection, Stream and Filter

### Objectives:

After finishing the following exercises, trainees will:

- Understand and practice with Classes, Object, Inheritance, Encapsulation, Abstraction and Polymorphism.
- Understand and practice with Control-of-flow statements.
- Understand and practice with method overloading, method overriding.
- Understand and practice with Java Collection, Streams and Filters.

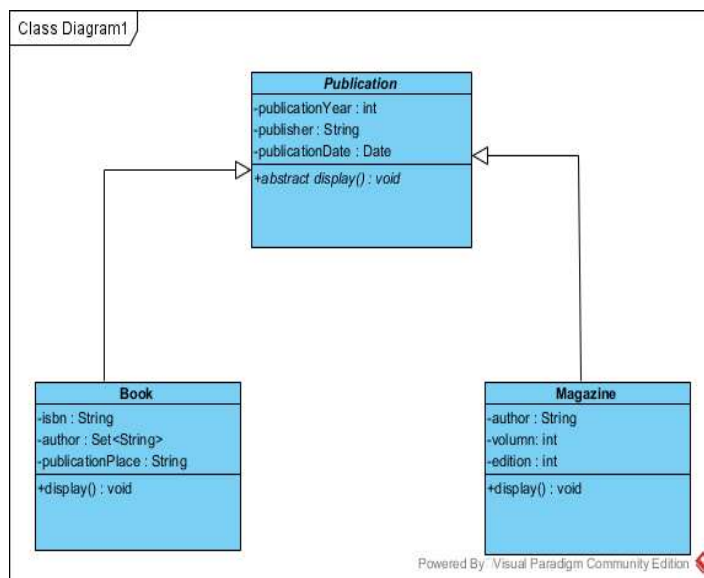
### Working Environments

- JDK 1.8
- Eclipse

### Assignment Descriptions

Create a Java console based application to manage publications in a library. In the library, there are two kinds of publication: book and magazine.

Refer to the class hierarchy which is described below, trainees need to create Java classes in order to implement the entities and the relationship between them.



**Publication** is an abstract superclass that contains the general information of book and magazine which will be extended. It includes *publication year*, *publisher* and *publication date*.

**Book** inherits from **Publication** class and has more attributes: *isbn* (international standard book number), *author* (a set of author name), *publication place*.

**Magazine** is similar to Book class. It inherits from Publication class and has more attributes: *author* (author name of magazine), *volume* and *edition*.

All of these classes must have getter/setter and constructor if needed.

You must override **display()** method in Book/Magazine class to print all of the information related to Book/Magazine.

### Validation Rules

- The *isbn* is a unique numeric commercial book identifier. The length of 'isbn' is in the 10-17 digit number and the '-' quote. Ex: 678-3-16-1486.

### Functional Requirements

1. The program must have a function to add a new book.
2. The program must have a function to add a new magazine.
3. The program must have a function to display the list of all books and magazines that have the same publication year and publisher.
4. The program must have a function to add an author to a specific book, if the author existed, the program should print a message "Author existed", otherwise print "Add successfully".
5. The program must have a function to display the list of top 10 magazines which have the largest volume.
6. The program must provide functions to:
  - search book by isbn
  - search book by author
  - search book by publisherSearch results should be sorted by isbn, publication date.

### User Interface Requirements

Create a new class named **LibraryManagement** that contains a main() method to display user interface.

The main screen allows user to select the following functions:

| Menu                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| =====<br>LIBRARY MANAGEMENT SYSTEM<br>=====<br>1. Add a book<br>2. Add a magazine<br>3. Display books and magazines<br>4. Add author to book<br>5. Display top 10 of magazines by volume<br>6. Search book by (isbn, author, publisher)<br><br>Please choose function you'd like to do: |

### Storage Data

- The user inputs data from the keyboard.
- Data is stored in the collection (List, Set...).
- Output data is displayed on the console.

**Guidelines:**

- Create a project named **JPL.L.A101**, create package ***fa.training.entities*** that contains classes/interfaces: **Publication, Book, Magazine**.
- Create other package ***fa.training.services*** that contains classes to implement Functional Requirements. The package ***fa.training.utils*** to implement Validation Rules requirements.

*Note, the functional requirements related to any entity, you have to create the service class corresponding to that entity. Ex: BookService, ...*

- Create a package named ***fa.training.main*** contains **LibraryManagement** class.

**-- THE END --**